

Logic in Computer Science, Prolog Test 2

BITS Pilani, K K Birla Goa Campus

Total: 10 Marks

Course No: *CS F214*

Time : 45 Minutes

November 13, 2025

Note:

- Store your file using your ID number. For example if your ID no is 2014A7PS1234G then your file name should be 2014A7PS1234G.pl
 - There should be no compilation error when you run your prolog program. Any error or nontermination, your program will not be evaluated.
 - Your file should contain predicates (mydescendant/2, myadd/3, mymult/3, mylistpartition/3, mysetpartition/3).
 - You have to provide definitions of all the auxiliary predicates used and you are not allowed to use any libraries.
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1. mydescendant(X,Y) returns true iff X is a descendant of Y. You can use the predicate parent(U,V) (here U is a parent of V and don't add any extra facts). We are assuming no one is descendant of himself/herself. (2 Marks)
2. Let us assume the non-negative integers are represented as the set $\{0, s(0), s(s(0)), \dots\}$. Write a definition in Prolog for myadd/3 where myadd(X,Y,Z) is true if $Z = X+Y$. Thus myadd(s(0),s(s(0)),s(s(s(0)))) is true, but myadd(s(0),s(s(0)),s(0)) is false. (2 Marks)
3. Now use myadd/3 definition to write a definition for mymult/3, where mymult(A,B,C) is true if $C = A \times B$ for non-negative integers A,B,C. (Hint: Use the fact the $A \times B$ is B added to itself A times). (2 Marks)

Your program definition for myadd(X,Y,Z) and mymult(X,Y,Z) should be able to correctly return an answer for the following kinds of queries: (i) Given X,Y find Z; (ii) given X,Z find Y; (iii) given Y,Z find X.

4. Assume sets are represented by lists. Let the Prolog predicate mylistpartition(A,B,C) be true for a given set A if B and C are disjoint subsets of A and $A = B \cup C$. For example mylistpartition([1,2,3],X,Y) will return $X=[], Y=[1,2,3]$; $X=[1], Y=[2,3]$; $X=[2], Y=[1,3]$; $X=[3], Y=[1,2]$; $X=[1,2], Y=[3]$; $X=[1,3], Y=[2]$; $X=[2,3], Y=[1]$; $X=[1,2,3], Y=[]$.

Write a definition for mylistpartition/3 which is correct when there are no duplicates in A.

Hint: $mylistpartition([? \mid ?], [? \mid ?], C) : -$
 $mylistpartition(?, ?, C).$

$mylistpartition([? \mid ?], B, [? \mid ?]) : -$
 $mylistpartition(?, B, ?).$

(2 Marks)

5. Lists do not correctly represent mathematical sets (elements can be duplicated). And because of this, mylistpartition may return wrong answers if there are duplicates in the first parameter. For example consider the answers for mylistpartition([1,2,3,1],B,C). Write a program for mysetpartition(A,B,C) such that B and C partitions the set A even if it has duplicates. Hint: make use of mylistpartition from the previous question. (2 Marks)

The queries for mylistpartition and mysetpartition won't have variables in the first parameter.

6. Bonus question

Suppose we have 5 propositions a–e that can be assigned values of true or false. Further assume we are interested in propositional formulae expressed in CNF (that is, a conjunctions of disjunctions of literals). Let a CNF formula be represented in Prolog by a list-of-lists. Thus the Prolog term “`[[a,b],[c],[not(d)]]`” represents the CNF formula $(a \vee b) \wedge (c) \wedge (\neg d)$. Here negated propositions are written using “not” (for example, “not(d)”). Write Prolog definition for the predicate `mysatisfied/1` that takes as input a CNF formula and returns true if there exists an interpretation for a–e that makes the formula true.

For example `mysatisfied([[a,b],[c],[not(d)]])` returns true and `mysatisfied([[a,b],[not(a)],[not(b)])` returns false.

Test queries for this problem won’t have variables.

Hint: Represent an interpretation in Prolog with predicate-symbol `interpretation/1`.

Thus, the Prolog fact “`interpretation([t,f,t,t,f]).`” represents the interpretation that a=true, b=false, c=true, d=true, e=false.

(5 Marks)